Question	Answer	Marks
8(a)	number of cycles per unit time	B1
8(b)(i)	period = $2\pi / 40\pi$ = 0.050 s = 50 ms	A1
8(b)(ii)	sinusoidal curve, starting at (0, 0) and initially increasing from there	B1
	periodic line showing 2 cycles with period 50 ms from t = 0 to t = 100 ms	B1
	all peaks shown at I = +3.5 A and all troughs shown at I = –3.5 A	B1
8(b)(iii)	$I_{r.m.s} = 3.5 / \sqrt{2}$ $= 2.5$ A	A1
8(c)	$P = I^2R$	C1
	peak power = $3.5^2 \times 680$ (= 8330 W) **or** mean power = $2.47^2 \times 680$ (= 4170 W)	M1
	peak and mean powers both calculated correctly, with supporting working, and compared leading to conclusion that mean power is half the peak power	A1

Question	Answer	Marks